

Ejercicios de regla de la cadena

Instrucción: Derive cada una de las siguientes funciones.

Ejercicios

1. $a(x) = e^{\sin(5x)}$
2. $b(x) = \ln(\cos(2x^2))$
3. $c(x) = e^{2x} \cos(x^2 + 1)$
4. $d(x) = \ln(x^2 + 3) \cdot \sin^2(4x)$
5. $e(x) = \frac{\cos^3(2x)}{\sqrt{x^2 + 1}}$
6. $f(x) = \frac{\ln(x^2 + 1)}{e^{2x}}$
7. $g(x) = \arcsin(x^2) \cdot e^{\sin(3x)}$
8. $h(x) = \arctan(\ln x) \cdot \cos^2(\sqrt{x})$
9. $i(x) = \arctan(e^{x^2}) \cdot \sin^2(\sqrt{x^2 + 1})$
10. $j(x) = \ln(\arctan(x^2)) \cdot e^{-\cos^2 x}$
11. $k(x) = \frac{\arcsin(x^2) \cdot \sin^3(e^x)}{\ln(x^2 + 1)}$
12. $l(x) = \frac{e^{-x^2} \cdot \arctan(\sin^2 x)}{\sqrt{1 + \cos^2 x}}$

Respuestas

1. $a'(x) = 5 \cos(5x) e^{\sin(5x)}$
2. $b'(x) = -4x \tan(2x^2)$
3. $c'(x) = 2e^{2x} [\cos(x^2 + 1) - x \sin(x^2 + 1)]$
4. $d'(x) = \frac{2x \sin^2(4x)}{x^2 + 3} + 4 \ln(x^2 + 3) \sin(8x)$
5. $e'(x) = -\frac{\cos^2(2x) [6(x^2 + 1) \sin(2x) + x \cos(2x)]}{(x^2 + 1)^{3/2}}$
6. $f'(x) = \frac{2[x - (x^2 + 1) \ln(x^2 + 1)]}{(x^2 + 1) e^{2x}}$
7. $g'(x) = e^{\sin(3x)} \left[\frac{2x}{\sqrt{1 - x^4}} + 3 \cos(3x) \arcsin(x^2) \right]$
8. $h'(x) = \frac{\cos^2(\sqrt{x})}{x(1 + \ln^2 x)} - \frac{\arctan(\ln x) \sin(\sqrt{x}) \cos(\sqrt{x})}{\sqrt{x}}$
9. $i'(x) = \frac{2x e^{x^2} \sin^2(\sqrt{x^2 + 1})}{1 + e^{2x^2}} + \frac{x \arctan(e^{x^2}) \sin(2\sqrt{x^2 + 1})}{\sqrt{x^2 + 1}}$
10. $j'(x) = e^{-\cos^2 x} \left[\frac{2x}{(1 + x^4) \arctan(x^2)} + \sin(2x) \ln(\arctan(x^2)) \right]$
11.
$$k'(x) = \frac{\sin^2(e^x) \left[\frac{2x \sin(e^x)}{\sqrt{1 - x^4}} + 3e^x \cos(e^x) \arcsin(x^2) \right] \ln(x^2 + 1)}{\ln^2(x^2 + 1)} - \frac{\frac{2x}{x^2 + 1} \arcsin(x^2) \sin^3(e^x)}{\ln^2(x^2 + 1)}$$
12. $l'(x) = \frac{e^{-x^2} \left\{ (1 + \cos^2 x) \left[\frac{\sin(2x)}{1 + \sin^4 x} - 2x \arctan(\sin^2 x) \right] + \frac{1}{2} \sin(2x) \arctan(\sin^2 x) \right\}}{(1 + \cos^2 x)^{3/2}}$